2201651- Experiment 6 : Study of Sample EDA from Kaggle notebooks

1.Describe all three data set in terms of following : how many features, their categories (num,text,categorical, nominal,ordinal etc), number of tuples, missing values.

Description of All Three Datasets

Air Pollution Dataset (World Air Quality Index)

- Number of Features: 14
- Feature Categories:
    - Categorical (Nominal/Text):
        - Country, City, AQI Category, CO AQI Category, Ozone AQI Category, NO2 AQI Category, PM2.5 AQI Category
    - Numerical (Integer):
        - AQI Value, CO AQI Value, Ozone AQI Value, NO2 AQI Value, PM2.5 AQI Value
    - Numerical (Float):
        - lat, lng
- Number of Tuples (Rows): 16,695
- Missing Values: Yes, some pollutant AQI values and categories may have missing entries depending on data completeness for each city.

IRIS Dataset

- Number of Features: 5
    - 4 numeric (predictive attributes)
    - 1 categorical (target class)
- Feature Categories:
    - Numerical (Continuous):
        - Sepal length (cm), Sepal width (cm), Petal length (cm), Petal width (cm)
    - Categorical (Nominal):
        - Class (Iris-setosa, Iris-versicolour, Iris-virginica)
- Number of Tuples (Instances): 150
    - 50 samples from each of the three classes

Wine Quality Dataset

- Number of Features: 12 (11 input features + 1 target)
- Feature Categories:
    - Numerical:
        - Fixed acidity, Volatile acidity, Citric acid, Residual sugar, Chlorides, Free sulfur dioxide, Total sulfur dioxide, Density, pH, Sulphates, Alcohol
    - Ordinal:
        - Quality (target score from 0–10, ordered)
- Number of Tuples: 4898
- Missing Values: No missing values
- Missing Values: None

2. In IRIS dataset, which non-graphical EDA is performed?

In the IRIS dataset, the following non-graphical Exploratory Data Analysis (EDA) techniques are performed:

- .head() and .tail() – to preview the first and last few records of the dataset
- .info() – to check data types, column names, and non-null counts
- .describe() – to generate summary statistics like mean, std dev, min, max, etc.
- .isnull().sum() – to check for any missing values in the dataset
- .value_counts() – to count the number of records per class label (Species)
- Correlation Analysis (.corr()) – to identify relationships between numeric features
- Unique Value Checks – using .unique() and .nunique() for feature-level uniqueness

These non-graphical methods provide essential insights into data distribution, class balance, and inter-feature relationships without using any visual plots.

3. In Air Quality dataset, which graph is used to perform multivariate analysis?

In the Air Quality dataset, the graph used for multivariate analysis is the Seaborn pairplot.

- Pairplot helps visualize pairwise relationships between multiple numerical variables (like AQI Value, PM2.5, CO, NO2, Ozone, etc.).
- It shows scatter plots for feature combinations and histograms or density plots on the diagonal.
- It also highlights trends, clusters, and correlations between pollutants across different cities or categories.

This makes pairplot a powerful tool for multivariate EDA in this dataset.

4. In Wine Quality dataset, can you do non-graphical EDA? Explain your answer.

Yes, non-graphical EDA can definitely be performed on the Wine Quality dataset.

Explanation:

Non-graphical EDA involves using statistical and descriptive methods instead of visualizations to explore the data. In this dataset, the following non-graphical techniques can be applied:

- .info() – to view column types, non-null counts, and overall structure
- .describe() – to get summary statistics like mean, median, standard deviation, min, max, etc. for all numeric features
- .isnull().sum() – to check for any missing values
- .value_counts() – to analyze how frequently each wine quality score appears
- .corr() – to compute correlations between chemical properties (e.g., alcohol, acidity) and wine quality
- .nunique() and .unique() – to check feature variability and uniqueness

These methods help in understanding data distribution, central tendencies, feature relationships, and data completeness — all without any plots.

So, non-graphical EDA is not only possible but essential as a first step before any visualization or modeling.